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A Framework for Road Authorities to Assess Their Readiness to Support Connected and Automated Driving

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Abstract. Connected and Automated Driving (CAD) will bring disruption to individuals, economies, and society. Most forms of CAD require some level of support from the infrastructure for their safe operation, in particular communications. However, additional infrastructure services to support CAD could improve safety and robustness and bring further benefits such as increased capacity. However, the infrastructure requirements of vehicle Original Equipment Manufacturers (OEMs) are not always clear, and it is therefore difficult for National Road Authorities (NRA) to prepare future levels of support for CAD, given rapidly evolving technology and uncertain projections of future CAD demand. There is a need to articulate those requirements, bringing stakeholders together to formulate a structured approach, and a roadmap that will advance safe and smart roads that support CAD.

This paper presents the DiREC project (consortium partners: TRL, ARUP, TU Delft, VTT, VTI and FEHRL) funded by the CEDR Transnational Road Research Programme Call 2020 with funding provided by CEDR members of Belgium (Flanders), Denmark, Ireland, Israel, Netherlands, Norway, Sweden, Switzerland and the United Kingdom. DiREC is seeking to address the above challenge. The project has established a CAV-Readiness Framework (CRF) based on a level of service approach to understand the needs of CAD, and to define the infrastructure and services that NRAs could provide to support these needs.

Keywords: CAD · Digital Infrastructure · Framework

1 Introduction

Whilst Connected and Automated Driving (CAD) will bring new choices and capability to users of the road network, it is an area of technology which is likely to bring disruption and change to the design and operation of road network infrastructure. Most forms of

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CAD require some level of infrastructure support for their safe operation. Additional infrastructure and services to support CAD would have the potential to improve safety even further, and to bring other benefits such as increased capacity or reduced congestion. However, conventional road infrastructure is already costly, amounting to around 1–2% of GDP for OECD countries, and accommodating the needs of new CAD technologies may increase these costs. But the infrastructure requirements from OEMs are not always clear, and it is difficult for NRAs to predict and plan the future levels of support needed for CAD, given the rapidly evolving technology and uncertain projections of future CAD demand. There is a need for better dialogue between NRAs, OEMs and service providers to articulate those requirements and to define a roadmap and responsibilities for achieving safer and smarter roads through CAD.

DiREC proposes to establish a CAV-Readiness Framework (CRF) to support dialogue between NRAs, OEMs and service providers, based on a Level of Service approach. The CRF would define the needs of CAD in terms of physical and digital infrastructure, services, and operational policies and procedures that NRAs could provide to support them. The CRF would consider a wide range of components that influence the ability of the NRA to become a digital road operator, including machine readability of physical infrastructure, digital services, connectivity, and aspects such as governance of the infrastructure and services, and legal and regulatory requirements.

This paper summarises the results of the DiREC project.

2 Stakeholder Management

DiREC conducted a review of literature and held a series of meetings and workshops with NRAs, OEMs, telecoms and other service providers. The purpose of this review was to determine the attitudes of NRAs towards CAD, and the levels and types of support that they are willing to provide. This review and engagement identified many differences of opinion among NRAs on how and even whether they should support CAD. These differences were the result of many factors, including lack of engagement between NRAs and OEMs and hence lack of understanding of needs, NRAs' lack of visibility on the rate and extent of technological progress in vehicle systems and communications technologies, NRAs' lack of involvement in development of national and regional regulations, and uncertainty regarding the uptake of CAD at all its different levels. In general, many NRAs feel underequipped to understanding current technologies and future direction.

Based on the stakeholder management, the project proceeded to identify the as-is situation, gaps, and future trends in CAV and CAD as well as the types of support and services that NRAs are providing. In addition, the appetite of NRAs to invest in infrastructure and services to support CAD including digital twins were investigated alongside the views of OEMs and service providers as to what support would be useful and what the priority areas are. The subject areas tackled are infrastructure design and operation, infrastructure connectivity, data exchange, digital twins, legal and regulatory, impact of emergent technologies and costs and benefits.

As a result of this review process, DiREC developed a vision to empower NRAs with the tools and techniques to make measurable assessments of their investment decisions that will drive the adoption of CAVs on the road network. The associated mission was to deliver a CAV-Readiness Framework (CRF) for NRAs that supports current and future requirements of the network. The CRF can help NRAs to define the ways in which they wish to support CAD, specifically through assistance to the provision of Cooperative Intelligent Transport Systems (C-ITS) services. C-ITS services and use cases are well-defined at a conceptual level, although their implementation and standardisation are still evolving. NRAs can support the implementation and evolution of C-ITS services through a mix of physical, digital and communications infrastructure, operations, and inputs to standards development. The CRF allows NRAs to articulate the type and extent of support they wish to provide. It helps NRAs to prioritise that support through identification and analysis of costs and benefits. It also helps NRAs to develop roadmaps to enable them to plan for that support.

3 CAV-Readiness Framework (CRF)

3.1 Development of the Framework

DiREC structured the CRF around C-ITS Services and Use Cases as defined under the C-ROADS project. C-ROADS is a joint initiative of European Member States and road operators for testing and implementing C-ITS services, with a desire for cross-border harmonisation and interoperability.

Under C-ROADS, the deployment of C-ITS is seen as evolutionary, starting with less complex use cases ("Day-1 Services" encompassing messages about traffic jams, hazardous locations, road works and slow or stationary vehicles, as well as weather information and speed advice). "Day 2" and "Day 3+" services are being investigated in R&D projects. Hence the C-ITS Service and Use Case definitions [1] provided a firm basis on which to implement the CRF, both now and in the future. The CRF thus becomes a framework which can be used by NRAs to help assess their aspirations and readiness to support CAD, and to implement individual C-ITS services and use cases. It does this by (see Fig. 1):

- Defining the C-ITS services to be provided.
- Breaking those services down into use cases and enablers.
- Scoring the NRAs readiness, aspirations and high-level assessment of costs and impacts of each enabler to help plan and prioritise the NRA support for CAD.

In C-ITS terminology, a service is a clustering of use cases based on a common denominator, for example an objective such as awareness of road works. Services are also known as 'applications'. C-ITS services are identified as shown in Table 1.

In DiREC terms, and for the purposes of the CRF, use cases are described using a set of enablers. The 'enabler' is the lowest building block of the CRF tool. Some examples of enablers are shown in Table 2 along with their category which groups them under one of Physical, Operation, Digital, Connectivity, or Standard.

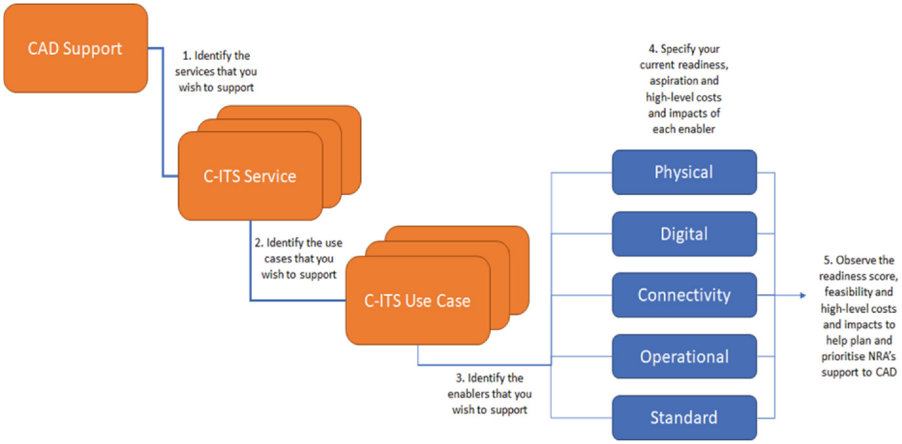

Fig. 1. Overview of the CRF

Table 1. C-ITS Services

C-ITS Services
In-Vehicle Signage (IVS)
Hazardous Location Notification (HLN)
Road Works Warning (RWW)
Signalized Intersections (SI)
Automated Vehicle Guidance (AVG)
Probe Vehicle Data (PVD)
In-Vehicle Signage (IVS)

Table 2. Sample enablers in the CRF tool

Enabler	Enable Category
Roadside Units (RSUs)	Physical
DENM messaging (ETSI EN 302 637–3)	Digital
ETSI EN 302 637–3	Standard
C-ITS Mobile Roadside ITS G5 System Profile	Connectivity
Cameras	Physical
Response plan	Operation

The CRF defines, for each enabler:

- the readiness of the NRA to provide or deploy each enabler individually.
- the aspiration of the NRA to provide or deploy each enabler.
- the feasibility threshold for the service which defines the minimum level of support provided by the NRA to make implementation of this enabler feasible.
- high-level costs and impacts of each enabler.

The scores, costs and impacts of each of the above can be rolled up to the level of the use case, and the service, and indeed the total package of support for CAD, to help plan and prioritise NRA support for each enabler.

3.2 Level of Service Approach

The Level of Service (LoS) is a widely employed metric that quantifies the performance and quality of a provided service, utilizing a predetermined scale. For C-ITS services (i.e., information provision) the aim is to provide information that is, among other things, accurate and timely to CAVs so they can react accordingly to events on the road network. The CRF aims to illustrate the progress of (NRAs) towards becoming a digital authority, meaning that the NRAs should provide information (data provision) to CAVs that are precise, accurate, and timely. As such, three distinct LoS categories can be defined:

- Basic: Minimum acceptable performance/quality.
- Enhanced: Not optimal but sufficient performance/quality.
- Advanced: Ideal or best performance/quality.

Additionally, a fourth level of service can be considered, for services that are already available. In this case, the LoS metric may be utilized by NRAs to assess the performance and quality of existing services.

The CRF LoS is a quality and performance evaluation metric based on a set of enablers with predefined requirements. The goal in the CRF was to develop a generic LoS tool that could be applied to any road environment, to any use case or technology. The definition should therefore be generic, applicable to all cases, and technology agnostic.

3.3 CRF Spreadsheet Tool

DiREC produced a spreadsheet version (CRF tool) of the CRF to demonstrate how the CRF could be applied in practice by an NRA.

The CRF tool is structured into separate tabs:

- Instructions for use of the tool, including colour coding of pages and fields to identify data inputs and outputs.
- Definition of which C-ITS services and use cases are to be analysed.
- Detailed analysis of a particular C-ITS service and use case to calculate the aspiration, readiness and feasibility of NRA support for that service and use case.
- Side-by-side comparison of the analysis of multiple services and use cases.
- Overall assessment of an NRA's readiness to implement an entire service.

This spreadsheet can be found on the DiREC website at: https://direcproject.com.

4 Conclusions and Recommendations

4.1 Using the CRF to Develop a Roadmap for an NRA

The CRF provides a framework to help NRAs understand their current readiness to provide or deploy C-ITS services and to understand potential investment decisions and link them to an overall strategic approach to deployment and delivery of a range of services. In addition to measuring the readiness of the NRA to support individual services and use cases, it also adds the concepts of the aspiration of the NRA to provide or deploy each enabler and helps identify a feasibility threshold for the service which defines the minimum level of support provided by the NRA to make implementation of this use case feasible.

4.2 Implementing and Refining the CRF

C-ITS and CAD are rapidly developing fields and to reflect this the CRF itself should evolve to refine its features and capabilities to help NRAs plan and prioritise their support to CAD. Figure 2 outlines a timeline of sets of potential actions that could help to implement and further develop the capabilities of the CRF. Some of the items proposed under the Actions are described below.

Actions 1: a) to support the effective use of the CRF, NRAs should undertake a number of internal workshops to help articulate their position on the topics raised; b) consolidate at NRAs, the list of enablers and the impacts associated with the various investment decisions.
Actions 2: a) Utilise existing asset management tools and Digital Twins in road operators to help consolidate the various equipment types, utilisation, impacts, and costs and benefits linked to the CRF and the deployment of CAD on the road network; b) develop a database at a National and European level to help inform the various parameters linked to the CRF and the wider Mobility sector.
Actions 3: a) Create a European CRF approach to help consolidate investments linked to the European Directives and local modification to allow for national investment decisions to take place; b) Link ongoing Road Investment decisions to the utilisation of the CRF and help articulate the business cases of the wider road network in this way.

Fig. 2. Action plan for implementation and development of the CRF

Reference

1. C-ROADS: C-ITS Service and Use Case Definitions (2022)